

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 5-00-153

WASTE DISCHARGE REQUIREMENTS
FOR
CITY OF MARICOPA
AND
UNITED STATES BUREAU OF LAND MANAGEMENT
MARICOPA WASTEWATER DISPOSAL FACILITY
KERN COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. The City of Maricopa (City) operates a wastewater disposal facility (Facility) on a 5.77-acre parcel of land leased from the United States Bureau of Land Management (BLM). The City and BLM are hereafter jointly referred as Discharger, with relative responsibilities further defined in Provision E.4. The Facility is in Section 7, T11N, R23W, SBB&M, as shown in Attachment A, which is part of this Order by reference.
2. The Facility features two disposal ponds covering about 2.25 acres total near the City, which is about 38 miles southwest of the City of Bakersfield. Currently, raw sewage is directly discharged to the disposal ponds without prior treatment. Disposal is by evaporation and percolation. The City reportedly intends to install an aerator in one of the disposal ponds. Accumulated sludge has reportedly been removed once from each pond within the past 15 years. The sludge was tested and reportedly disked into nearby soils.
3. Waste Discharge Requirements Order No. 76-252, adopted by the Board on 25 February 1976, prescribes requirements for the discharge to the Facility of up to 30,000 gallons per day (gpd). Order No. 76-252 is not consistent with current plans and policies of the Board, and does not reflect the Facility's current operations.
4. The Facility serves 275 homes, 10 businesses and a school. The Facility's influent flows had not been measured prior to February 2000, when the City temporarily installed a flow meter at the Facility for 16 days. Flows entering the Facility during this period ranged from 25,870 to 68,790 gallons per day (gpd), and averaged around 45,600 gpd.
5. The City has reportedly experienced little growth within the past decade and is not expected to experience any significant growth in the near future. The Facility has not experienced any capacity problems to date.

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6. The City is purchasing the 15.8-acre parcel from BLM on which the Facility is located. The City reports that the process of acquiring the property will take 12 to 18 months to complete and that it plans to use the land to eventually expand the Facility's disposal capacity.
7. The general area has experienced significant subsidence due to the region's oil extraction activity, according to BLM. In particular, the east side of disposal pond 2 has settled one foot relative to the rest of the disposal area.
8. Average annual rainfall in the area is 6.5 inches and annual average pan evaporation rate is 63 inches.
9. Area soils consist of a recent alluvial fan deposit composed primarily of laterally discontinuous lenses of low permeability silts, sands, and clays of Quaternary age. The deposit is approximately 300 feet thick and overlies the Tulare formation.
10. Area groundwater is first encountered at an approximate depth of 200 feet below ground surface and is of poor mineral quality with EC (conductivity at 25°C) of 4,000 to 6,000 µmhos/cm and total dissolved solids greater than 4,000 mg/l.
11. Source water for residential and commercial use is groundwater supplied by West Kern County Water District and is pumped from wells near the Kern River and Elk Hills. Recent water supply reports from West Kern County Water District indicate the source water has an EC of about 370 µmhos/cm, total dissolved solids of about 290 mg/l, and nitrate-nitrogen of about 1.0 mg/l.
12. Due to the poor quality of area groundwater and to the need to import potable water to the area, there is little if any agricultural land use in the general vicinity.
13. The Facility is within the Taft Hydrologic Area (No. 557.20) in the South Valley Floor Hydrologic Unit, as depicted on interagency hydrologic maps prepared by the California Department of Water Resources in August 1986. Surface water drainage is to Bitterwater Creek.
14. The *Water Quality Control Plan for the Tulare Lake Basin, Second Edition*, (hereafter Basin Plan) designates beneficial uses, establishes water quality objectives, and contains implementation plans and policies for all waters of the Basin. These requirements implement the Basin Plan
15. From its origin in the coastal mountains, Bitterwater Creek flows east towards the floor of the southern San Joaquin Valley. The beneficial uses of the West Side Streams and Valley Floor Water, as identified in the Basin Plan, include agricultural supply; industrial supply; hydropower generation; water contact and noncontact water recreation; warm fresh water habitat; wildlife habitat, rare, threatened, or endangered species; and groundwater recharge.

16. The Basin Plan identifies the beneficial uses of area groundwater as municipal and industrial supply. As described in Finding No. 10, the EC of area groundwater ranges from 4,000 to 6,000 $\mu\text{mhos/cm}$. According to State Water Resources Control Board Resolution No. 88-63, *Adoption of Policy Entitled "Source of Drinking Water,"* groundwaters with EC concentrations exceeding 5,000 $\mu\text{mhos/cm}$ may be considered by a regional board as unsuitable for municipal or domestic water supply. Area groundwater does not currently supply a public water system, nor is it reasonably expected to do so in the future. As described in Finding No. 11, all source water for municipal uses is imported.
17. According to the Basin Plan, evaporation of reclaimable wastewater will not be an acceptable permanent disposal method where opportunity exists to replace an existing use or proposed use of fresh water with reclaimed water. As described in Finding No. 12, the area's land use does not include agricultural operations that could readily offer opportunities for water reclamation.
18. The Basin Plan requires at least a primary level of treatment for isolated domestic wastewater facilities with relatively minor discharges to land where there is little risk of nuisance or water quality degradation. The Facility is in a remote, sparsely populated area that is not used for agriculture or housing. The potential of waste constituents in the discharge to degrade area groundwater is minimal.
19. The Board has considered antidegradation pursuant to State Water Resources Control Board Resolution No. 68-16 and finds that the permitted discharge is consistent with those provisions. The discharge will not unreasonably affect present and anticipated use of underlying groundwater as a source of industrial supply. If the Board determines that the discharge causes nuisance or water quality degradation, this Order will be reopened for consideration the addition of numerical effluent limits that would, in effect, require the City to provide a higher level of wastewater treatment.
20. The action to update waste discharge requirements for this existing facility is exempt from the provisions of the California Environmental Quality Act (CEQA) in accordance with Title 14, CCR, Section 15301. This Order allows for a higher discharge flow rate than that previously permitted. However, the increase is not due to any expansion but reflects current flows to the existing facility.
21. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
22. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Waste Discharge Requirements Order No. 76-252 is rescinded and the City of Maricopa and United States Bureau of Land Management, as well as their agents, successors,

and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

A. Discharge Prohibitions

1. Discharge of waste to surface waters or surface water drainage courses is prohibited.
2. Overflow of waste is prohibited.
3. Discharge of waste classified as 'hazardous,' as defined in Section 2521 (a) of Title 23, CCR, Section 2510 et seq., or 'designated,' as defined in Section 13173 of the California Water Code, is prohibited.

B. Discharge Specifications

1. The monthly average discharge shall not exceed the calculated capacity of the Facility's disposal ponds, as determined by the technical report required pursuant to Provision E.3.
2. All influent and effluent must be maintained within the designated disposal area.
3. Objectionable odors originating at the Facility shall not be perceivable beyond the limits of the designated disposal area.
4. Ponds shall not have a pH less than 6 or greater than 9.
5. Discharge EC (conductivity at 25°C) shall not exceed source water EC plus 500 µmhos/cm.
6. Ponds shall be managed to prevent breeding of mosquitoes. In particular,
 - a. An erosion control program should assure that small coves and irregularities are not created around the perimeter of the water surface.
 - b. Weeds shall be minimized through control of water depth, harvesting, and herbicides.
 - c. Dead algae, vegetation, and debris shall not accumulate on the water surface.
7. Public contact with wastewater shall be precluded through such means as fences and signs, or acceptable alternatives.
8. The Facility's disposal ponds shall have sufficient capacity to accommodate wastewater flow at design seasonal precipitation while meeting all terms of this Order. Design seasonal precipitation shall be based on total annual precipitation using a return period of 100 years,

distributed monthly in accordance with historical rainfall patterns. Freeboard shall never be less than two feet (measured vertically to the lowest point of potential overflow).

9. On or about **1 October** of each year, available disposal pond storage capacity shall at least equal the volume necessary to comply with Discharge Specification B.8.
10. Solids, scum, grease and other wastewater materials shall not be visible on the surface of the disposal ponds.

C. Sludge Disposal Specifications

1. Collected sludges, scums, and other solids removed from liquid wastes shall be disposed of in a manner approved by the Executive Officer and consistent with *Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste*, as set forth in Title 27, CCR, Division 2, Subdivision 1, Section 20005 et seq.
2. Any proposed change in sludge use or disposal practice shall be reported to the Executive Officer and USEPA Regional Administrator **at least 90 days in advance** of the change.

D. Groundwater Limitations

The discharge, in combination with other sources, shall not cause underlying groundwater to contain waste constituents in concentrations statistically greater than background water quality, except of conductivity. Regarding EC, the Discharger shall not cause the groundwater to exceed an incremental increase in EC greater than 25 µmhos/cm over any five-year period.

E. Provisions

1. The Discharger shall comply with Monitoring and Reporting Program No. 5-00-153, which is part of this Order, and any revisions thereto as ordered by the Executive Officer.
2. The Discharger shall comply with *Standard Provisions and Reporting Requirements for Waste Discharge Requirements*, dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as *Standard Provision(s)*.
3. By **31 December 2000**, the City shall submit a written technical report with data, calculations, assumptions, and other information determining the average monthly discharge rate that assures compliance with Discharge Specification B.8. The technical report shall be prepared and certified by a California registered civil engineer experienced in wastewater treatment and disposal facility design. Upon written approval of the technical report by the Executive Officer, this Provision will be considered satisfied.

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4. The City of Maricopa shall retain primary responsibility for the day-to-day compliance with this Order. The United States Bureau of Land Management, as owner of the disposal property, shall have the secondary responsibility until such time that the City attains ownership from BLM of the property described in Finding No. 6.
5. In the event of any change in control or ownership of land or waste discharge facilities described herein, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.

To assume operation under this Order, the succeeding owner or operator must apply in writing to the Executive Officer requesting transfer of the Order. The request must contain the requesting entity's full legal name, the state of incorporation if a corporation, the name and address and telephone number of the persons responsible for contact with the Board, and a statement. The statement shall comply with the signatory paragraph of Standard Provision B.3 and state that the new owner or operator assumes full responsibility for compliance with this Order. Failure to submit the request shall be considered a discharge without requirements, a violation of the California Water Code. Transfer shall be approved or disapproved by the Executive Officer.

6. The Discharger must comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.
7. A copy of this Order shall be kept at the Facility for reference by the personnel responsible for wastewater management. These persons shall be familiar with its contents.
8. The Board will review this Order periodically and will revise requirements when necessary.

I, GARY M. CARLTON, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 16 June 2000.

GARY M. CARLTON, Executive Officer

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. 5-00-153

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Specific sample station locations shall be established with concurrence of the Board's staff, and a description of the stations shall be submitted to the Board and attached to this Program.

DISPOSAL POND MONITORING

The freeboard shall be monitored on all storage ponds to the nearest tenth of a foot. A permanent marker shall be placed in each pond with calibration including the water level at design capacity and available operational freeboard. Monitoring of the disposal ponds shall include at least the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Frequency</u>
Flow	gpd	Estimate	Monthly
BOD ¹	mg/l	Grab	Monthly
pH	pH Units	Grab	Monthly
EC ²	µmhos/cm	Grab	Monthly
<u>Freeboard</u>	feet	Observation	Monthly

1 Five-day biochemical oxygen demand
2 Conductivity at 25°C)

In addition, the Discharger shall inspect the condition of the disposal ponds once per week and write visual observations in a bound log book. Notations shall include observations of whether weeds are developing in the water or along the bank, and their location; whether dead algae, vegetation, scum, or debris are accumulating on the pond surface and their location; whether burrowing animals or insects are present; and the color of the ponds (e.g., dark sparkling green, dull green, yellow, gray, tan, brown, etc.). A copy of the entries made in the log during each month shall be submitted along with the monitoring report the following month. Where the operation and maintenance manual indicates remedial action is necessary, the Discharger shall briefly explain in the transmittal what action was taken or is scheduled to be taken.

SLUDGE MONITORING

When sludge is removed from the disposal ponds, but prior to disposal, a composite sample shall be collected and analyzed on a dry weight basis for Total Solids (%), nitrogen (total, $\text{NH}_4\text{-N}$, and $\text{NO}_3\text{-N}$), total phosphorus, total potassium, and totals of specific metals (i.e., Pb, Zn, Cu, Ni, Cd, and Ag). Analysis of soluble concentration of these specific metal shall also be included. Analytical results shall be submitted to the Executive Officer within 60 days of sludge removal. If final disposal is by land application, a technical report analyzing application rates and procedures relative to California Department of Health Services' *Process Design Manual for Land Application of Municipal Sludges* and Title 23, California Code of Regulations, Section 2511(f), shall be completed and submitted to the Executive Officer for approval. The report shall be prepared by a California registered civil engineer experienced in wastewater treatment and disposal facility design.

REPORTING

Monthly monitoring data shall be reported in monthly monitoring reports. Monthly reports shall be submitted to the Board by the **1st day of the second month following sampling**. Annual monitoring reports shall be submitted by **1 February of each year**.

In reporting the monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in such a manner that illustrates clearly whether the Discharger complies with waste discharge requirements.

If the Discharger monitors any pollutant at the locations designated herein more frequently than is required by this Order, the results of such monitoring shall be included in the discharge monitoring report.

The Discharger may also be requested to submit an annual report to the Board with tabular and graphical summaries of the monitoring data obtained during the previous year. Any such request shall be made in writing. The report shall discuss any corrective actions taken or planned that may be needed to bring the discharge into full compliance with the waste discharge requirements.

By **1 February of each year**, the Discharger shall submit a written report to the Executive Officer containing the following:

- a. The names, certificate grades, and general responsibilities of all persons in charge of wastewater treatment and disposal.
- b. The names and telephone numbers of persons to contact regarding wastewater disposal for emergency and routine situations.
- c. The most recent annual water quality report prepared by West Kern Water District for the City of Maricopa municipal drinking water supply.
- d. The results of an annual evaluation conducted pursuant to Standard Provision E.4.

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All reports submitted in response to this Order shall comply with the signatory requirements in Standard Provision B.3.

The Discharger shall implement the above monitoring program on the first day of the month following adoption of this Order.

Ordered by: _____
GARY M. CARLTON, Executive Officer

16 June 2000
(Date)

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INFORMATION SHEET

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The City of Maricopa operates a wastewater disposal facility (Facility) on a 5.77-acre parcel one mile east of Maricopa on Road 166 and along a dirt road, 0.25 mile north of Road 166. The land on which the Facility is located is leased from the United States Bureau of Land Management (BLM). The City has applied with BLM to purchase 15.8 acres on which the Facility is located. The City has indicated that the parcel acquired from BLM may be used for future expansion of the Facility and that it expects to gain legal ownership of the property within 12 to 18 months.

The Facility currently serves 10 businesses, one school, and 275 households. The remaining households in the City are served by septic systems. The discharge is currently regulated by Waste Discharge Requirements Order No. 76-252, which limits the discharge flow to 30,000 gallons per day (gpd). Until recently, the Facility's discharge flow was not metered, but estimated. During a 16-day period in February 2000, the City measured the flows entering the Facility as averaging about 45,600 gallons per day. The current discharge flow rate, while in excess of that prescribed by Order No. 76-252, apparently has not caused any capacity problems for the Facility. The City has reportedly experienced little growth during the past decade and reportedly does not expect much growth in the near future.

The influent is primarily domestic waste which, transported via sewer lines, is discharged into one of two disposal ponds without any primary or secondary treatment. The City reportedly intends to operate one pond as an aerated pond, and plans to install an aerator as soon as possible. The disposal ponds encompass 2.25 acres and the Facility is fenced and posted with warning signs.

Oil extraction activity around Maricopa has caused significant subsidence in the area, according to BLM. Therefore, the Order includes a provision that all wastewater must be maintained within the designated disposal area at all times. Any pond failure caused by subsidence would be a violation of this Order and the City must take all necessary steps to avoid such occurrence.

Average annual rainfall in the area is 6.5 inches and annual average pan evaporation rate is 63 inches. Soils at the disposal area consist of a recent alluvial fan deposit composed primarily of laterally discontinuous lenses of low permeability silts, sands, and clays. Area groundwater is first encountered at about 200 feet below ground surface and is of poor mineral quality. Source water for residential and commercial use is imported from West Kern County Water District. Due to the poor quality of area groundwater and to the need to import potable water to the area, there is little if any agricultural land use in the general vicinity.

The ponds have the potential to create odor nuisance conditions. Therefore, the Order requires that no objectionable odors from the ponds should be perceivable beyond the disposal area. Beyond this area the public is present and might be affected. To operate without problems, the ponds should have near

neutral pH and normal algae growth. A pH less than 6.5 indicates acidic conditions and greater than 8.5 indicates basic conditions, too extreme for effective operation. The Order requires that the ponded water have a pH between 6 and 9.

Ponded water creates a potential for mosquitoes to breed and create vector nuisances. The Order has requirements on pond maintenance such that this threat is minimized. Also, in the interest of preventing overflow conditions or failure of bank embankments, the Order requires the ponds to have adequate capacity with a minimum of two feet of freeboard.

This Order includes groundwater limitations based on the Basin Plan.

The City has removed sludge once from each pond during the past 15 years. Historically, the sludge was tested and disposed of on-site. The Order requires the City to monitor sludge quality prior to disposal.

The permitted discharge is consistent with the antidegradation provisions in State Water Resources Control Board Resolution No. 68-16. The discharge will not unreasonably affect present and anticipated use of underlying groundwater as a source of industrial supply. If the Board determines that the discharge causes nuisance or water quality degradation, this Order will be reopened for consideration the addition of numerical effluent limits that would, in effect, require the City to provide a higher level of wastewater treatment.

This is an existing facility and the action to update requirements is exempt from the provisions of the California Environmental Quality Act in accordance with Title 14, California Code of Regulations, Section 15301. This Order allows for a higher discharge flow rate than that previously permitted. However, the increase is not due to any expansion but reflects current flows to the existing facility.

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